HL-CYB-001

T3 report - PDF documentation and folder organization

Enterprise Zero Trust Security Architecture

Document version e5542bbdef11

Report state: FINALIZED

Contents

1. A. Task Information 3

2. B. Implementation Summary 3

3. C. Testing and Validation 4

4. D. Security and Quality Review 5

5. E. GitHub Operations 5

6. F. Outstanding Issues 6

7. G. Completion Status 6

8. H. Next Task 6

Report state: FINALIZED

Finalization date: 2026-11-05, America/Toronto. Evidence cutoff: implementation commit `53d106234cf7c0128e70d395fd8791628b7762e5` and its completed push/PR checks. This report's final PDF, publication commit, exact-head CI, merge, and synchronization are verified after freezing this record and reported in the final task response; they are not predicted here.

1. A. Task Information

Table 1. A. Task Information

Field	Record
Task ID	T3
Title	PDF documentation and folder organization
Execution date	2026-11-05, America/Toronto
Scope and objectives	Organize editable documents by purpose, generate faithful professional PDFs, implement a reproducible Windows build and CI checks, preserve historical sources, and publish through the normal PR workflow
Dependencies	Owner PDF/folder addendum; completed T2/reporting governance; inspected free PDF toolchain; authenticated repository access; no infrastructure approval
Branch / baseline	`task/003-pdf-documentation`; `ab0a96e3c4dc6cb113818ee3b57ee4618a0d8cc8`

2. B. Implementation Summary

Moved eleven editable sources into purpose folders: assignment to `docs/assignments/`, lab guide to `docs/lab-guides/`, architecture to `docs/architecture/`, security to `docs/security/`, test plan/acceptance to `docs/testing/`, decisions/risks/references to `docs/governance/`, the final template to `docs/final-report/`, and glossary to `docs/guides/`. Updated current navigation, README, AGENTS.md, the report index, validator paths, and CI. Two small decision/risk compatibility pages preserve links in immutable reports; they are not duplicate source editions.

Preserved finalized T1, its addendum, T2, and bootstrap evidence byte-for-byte. Generated twenty-four adjacent PDFs: thirteen documentation/library/build editions, five report/index editions, two root guides, and four folder/evidence editions. The documentation library and generated manifest identify each source/PDF pair. The project final report remains visibly a template, with no invented implementation results.

Used inspected bundled Python 3.12.14, ReportLab 4.4.9, pypdf 6.10.0, and Poppler 26.07.0. Added only Mistune 3.4.0/Pygments 2.21.0 to an ignored project-local dependency folder; pinned the full documentation dependency set for Windows/Linux. Added `scripts/build_docs.py`, `tests/test_build_docs.py`, `requirements-docs.txt`, the generated manifest, library/build instructions, generated PDFs, and this report. Updated `.gitattributes` for stable text/binary handling and `.gitignore` for temporary previews/local environments. CI installs only documentation dependencies and checks fresh regeneration.

The deterministic renderer embeds fonts, adds source-derived document versions, numbered headings, title/contents pages where appropriate, repeatable table headers, highlighted/wrapped code, vector architecture, page numbers, clickable references, and named heading destinations. All twelve planned stages gained explicit Objective, Required tools, and Troubleshooting fields alongside their existing procedures/checks/rollback. No procedure was executed. No infrastructure, production, account privacy,

project license, or global software setting changed.

3. C. Testing and Validation

Table 2. C. Testing and Validation

Check	Expected result	Observed result	Status
Source migration and internal links	Content/IDs/commands preserved; all current and historical navigation resolves	Eleven moves accounted for; all links passed; sixteen requirements and the rubric/control/architecture tables retained; eight original lab code blocks unchanged; twelve stages have all ten required fields	PASS
Frozen report/evidence comparisons	All finalized historical bytes unchanged	Exact Git-baseline comparisons passed for all four finalized/historical Markdown artifacts	PASS
PDF batch build	Every approved source has an adjacent readable/searchable PDF	Twenty-four source/PDF pairs generated and opened; every PDF has extractable/searchable text	PASS
PDF source/structure/navigation check	Correct source hashes, pages, text coverage, links, bookmarks, and footers	Published implementation snapshot: 125 pages, 100% normalized source-token coverage, 193 per-document URI links, 309 named heading destinations, valid footers and no orphan headings	PASS
Deterministic rebuild	Fresh render matches existing output and no-change build causes no churn	Full fresh in-memory rebuild matched bytes; no-change build skipped all twenty-four outputs; an isolated table-column improvement changed only its two affected PDFs	PASS
Representative visual inspection	Readable title/TOC/tables/code/diagram/page breaks; no clipping/overlap/missing glyphs	Viewed all twenty-four first pages and eleven detailed samples; corrected empty-page/orphan/narrow-column issues; final character-bound checks found zero violations across the snapshot	PASS
Focused tests and publication checks	Actual test results recorded; new public artifacts clean	All twenty-five tests passed; generated PDF text/metadata and source scans had zero selected new-artifact findings; unknown/drifted PDFs rejected	PASS
Implementation GitHub CI	Hosted source/PDF regeneration, publication hygiene, tests, and whitespace pass for the implementation head	Push run 42150291539 and PR run 42150472267 completed successfully for `53d106234cf7c0128e70d395fd8791628b7762e5`	PASS
Final report publication/merge and synchronization	Exact-head checks/requirements pass; normal noreply merge; matching main refs	Follows this immutable report's evidence cutoff; final response must record actual results	NOT TESTED

4. D. Security and Quality Review

PDFs must be generated from reviewed public-safe Markdown without raw evidence, actual addresses, secrets, private filesystem paths, or executable content. New binary publication is restricted to source-matched manifest-listed PDFs. The project final report remains a template with no fabricated implementation or acceptance.

Executed `scripts/build_docs.py`, `scripts/build_docs.py --check --rebuild-check`, `scripts/validate_docs.py`, `python -m unittest discover -s tests -v` with the documented dependency runtime, Git whitespace/source-history checks, Poppler page rendering, and independent pypdf/pdfplumber text/metadata/geometry inspection. Initial text/layout guards exposed split decision IDs and table pagination issues; ID widths, long-table grouping, and heading placement were corrected, then PDFs were rebuilt and inspected again. These were documentation defects, not lab/security findings. Code/diagram source and requirement/control tables were compared with the Git baseline rather than inferred from a build success flag.

Historical email metadata RK-17 remains an owner decision; no history rewrite is authorized. All operational lab/security/isolation tests remain NOT TESTED. Visual checks and text extraction are separate observations; neither implies owner UAT.

5. E. GitHub Operations

Table 3. E. GitHub Operations

Item	Observed state at evidence cutoff
Repository / branch	`waqashamdani-ca/enterprise-zero-trust-security`; `task/003-pdf-documentation`
Dependency PR	PR 2 merged at `ab0a96e3c4dc6cb113818ee3b57ee4618a0d8cc8`; both merge identity fields were noreply; local/remote main matched
Dependency merge CI	Run 42144602697: success
Authenticated identity / permission	`waqashamdani-ca`; repository API reports admin/push permission; default branch `main`
T3 implementation commit	`53d106234cf7c0128e70d395fd8791628b7762e5`; local branch and remote tracking branch agree
T3 pull request	PR 3, open with clean merge state; no required branch protections/rules observed
T3 implementation CI	Push run 42150291539 and PR run 42150472267: success; hosted steps include fresh PDF regeneration and focused tests
Final report / merge	Publication of this final record and its regenerated PDF follows the cutoff; PR was not merged at the cutoff
Synchronization	Clean implementation branch at `53d106234cf7c0128e70d395fd8791628b7762e5`; fetched remote main remains `ab0a96e3c4dc6cb113818ee3b57ee4618a0d8cc8`; final main read-back pending publication

6. F. Outstanding Issues

Local build, artifact checks, tests, visual review, and implementation branch/PR CI are complete. Publication checks for the frozen report, merge, and final synchronization remain pending at this cutoff. The renderer supports the repository's reviewed Mermaid flowchart subset and fails on unsupported syntax; it is not a general Mermaid engine. External/named-destination behavior depends on the PDF reader. Typography normalizes unsupported arrow/dash glyphs without modifying Markdown. Visual samples and automated geometry checks are separate from owner acceptance.

Historical email metadata RK-17, prior external-reference transport limits, owner UAT, and future hardware/egress/backup/license/protection decisions remain unresolved. No new reachability or operational security claim is made. The final report's own PDF/hash/page count will be rebuilt and checked again after finalization; final publication results belong to the PR/final response after the stated evidence cutoff.

7. G. Completion Status

Table 4. G. Completion Status

Area	Status
Engineering	Authorized documentation/PDF implementation VALIDATED; finalized report publication/read-back follows the stated cutoff
Independent verification	Source comparisons, PDF extraction/render review, fresh rebuild, and hosted Ubuntu CI passed; no independent human reviewer sign-off claimed
Owner UAT	PENDING
Remaining owner decisions	Acceptance of documentation/PDFs; historical email privacy remediation; separate approval for any future lab work

8. H. Next Task

Stop after this documentation/PDF scope and request owner review. The next recommended bounded engineering task remains a read-only Stage A hardware and isolation readiness assessment. It requires separate explicit owner approval, confirmed asset access/ownership, and private evidence handling. No Proxmox installation, network configuration, VM creation, or infrastructure/security testing is authorized here.